Satvik Srivastava

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Education

Indian Institute of Information Technology, Kota **Nov 2022 – July 2026**
B.Tech in Electronics and Communication Engineering CGPA: 7.95/10

Experience

Xelron AI **Feb 2026 – May 2026**
SDE Intern Remote

- Designed realistic coding and debugging benchmark tasks for **Terminal-Bench 2.0** to evaluate AI agents in terminal-based software-engineering environments.
- Built each task end-to-end with task instructions, **Docker**-based environments, reference/**ORACLE** solutions, and automated verifiers to objectively test and validate whether an agent completed the required task.
- Submitted **67** tasks through the review pipeline, with **53** accepted.

Projects

Ytscraper | Python, FastAPI, Qdrant, Faster-Whisper, Groq API | GitHub | Demo **Mar 2026 – Apr 2026**

- Developed a **timestamp-level semantic search engine** over a **126-lecture (68.6 hrs) DSA YouTube playlist**, letting students ask doubts in **Hinglish/English** and jump to the exact second in an embedded player.
- Engineered an end-to-end **RAG** pipeline: **Whisper large-v3** batched transcription **(27 hrs to 8.5 hrs)**, custom time-window chunking into **2,933** chunks, **MiniLM** embeddings in **Qdrant**, and title-boost re-ranking that raised top-1 accuracy from **9/12** to **12/12**.
- Built a **FastAPI** backend with rate limiting and a **hallucination guard** (distance cutoff plus citation checks) that kept **20/20** in-syllabus questions and refused **10/10** off-topic ones, with an optional **Groq** explanation layer.

Resume Analyzer | React.js, Tailwind, Python, FastAPI, Groq, Pydantic | GitHub | Demo **Feb 2026 – Mar 2026**

- Built a **full-stack web application** where recruiters upload a **resume** and **job description** to receive a **0–100** match score, matching/missing skills, and a hiring verdict.
- Architected a **React** frontend integrated with a **FastAPI** backend, where an LLM **(Groq gpt-oss-120b)** extracts structured candidate and job data, **Pydantic** validates it, and deterministic Python rules calculate the score.
- Decoupled LLM-based understanding from deterministic scoring to make candidate evaluation consistent, explainable, and independent of subjective model-generated scores.

WanderLust | JavaScript, MongoDB, Express.js, Node.js, Passport.js | GitHub | Demo **Sept 2025 – Oct 2025**

- Delivered a **full-stack** property rental platform where users list properties with images and map locations, browse stays, and post reviews.
- Structured the application around **MVC** with **Node.js**, **Express.js**, **MongoDB Atlas (NoSQL)** and **EJS**, integrating **Passport.js** for authentication, **Cloudinary** for image storage, **Mapbox** for location geocoding, and **Joi** for request validation.
- Enforced multi-user access controls through owner-only listing modifications and review-author permissions, while cascading related listings and reviews during account deletion to preserve data consistency.

Technical Skills

- **Languages & Core CS:** C++, Python, JavaScript, Data Structures and Algorithms
- **Frontend:** HTML5, React.js, Tailwind CSS, Bootstrap, EJS
- **Backend & APIs:** Node.js, Express.js, FastAPI, REST APIs
- **Databases:** MongoDB, MySQL, QdrantDB
- **AI:** LLM Integration, RAG, Prompt Engineering, Groq, Whisper
- **Tools & Services:** Git, GitHub , Hoppscotch, Render, Cloudinary

Achievements

- Achieved **Specialist** rank on **Codeforces** with a highest rating of **1423**.
- Earned a **4 Star** rating on **CodeChef**, reaching a peak rating of **1828** and a global rank of **116**.
- Solved **1600+** DSA problems and reached **Knight** status on **LeetCode** with a maximum rating of **1987**.